

## Energy, Power &amp; KWH Meter Components Special :

- Latching Relays
- Current Transformers
- Shunt for KWH Meter
- Power Shunt for Power Meter
- European Type Counter
- Indian Type Counters
- Ultra & Super Capacitors
- IC ENERGY METERING 3 PHASE
- IC ENERGY METERING 1PHASE
- Evaluation Boards



From Analog Devices & Cirrus Logic  
From Analog Devices & Cirrus Logic

## Company Recognised Channel Partners &amp; INR Outlet :

**BOURNS®**

Helipot, Circuit Protection Devices,  
Triacs, Transistors, Bridge Rectifiers, Inductors

**Applications :** Solar Inverters, Led Drivers,  
Power Supplies, Energy Meters

**CIRCUIT LOGIC®**

A/D Convertors, Embedded Processors, Voltage &  
Sine Wave References, power / Operational  
& PWM Amplifiers, Energy Measurement & Energy  
Control Integrated Circuits.

## Our Strong Lines And Direct Sourcing:



## Why Amar Electronics:

- 100% warranted Products.
- CENVAT Credit Available.
- \*100% Traceability with Certificate of Conformance.
- Test Reports provided on Request.
- Competitive Price.
- Shortest Lead time.
- Multi Currency Invoice

**AMAR ELECTRONICS**

1192, Shukrawar Pet, Coimbatore 641001, Ph : 0422 - 2470320, 3298869  
Mob: 91 787 1110 302, Email: sales@amarelectronics.com

<http://www.amarelectronics.com>

## Key challenges

Despite the robust government backing, the market is hindered by high capital requirements and the risks posed by the lack for grid parity of large solar farms. Most developers faced issues with financial closure. There is an overall reluctance on the part of financial institutions to lend to solar projects. Banks are asking for large collaterals to the extent of over 110 per cent and are reluctant to finance purely on the basis of power purchase agreements.

In addition, the interest rates are also very high with some banks funding at rates as high as 15-16 per cent. Market participants will have to sort out the affordability issue if they desire wide-scale deployment of rural electrification programmes.

Policy and procedural delays are causing major problems in completion of land transfer and this has a domino effect, leading to delays in financial closure and consequently delays in commissioning of the project. In addition, there are delays in getting clearances, like the Pollution Clearance Certificate from the concerned agency. Besides, there is no clarity on the approvals required. For instance, few people know that projects under 5 MW do not require a pollution clearance certificate.

The country also lacks a well developed transmission and distribution (T&D) network for evacuation of solar power from remotely located plants.

These issues need to be addressed to ensure that the solar market in the country continues on its growth trajectory.

—Rajesh Menon, deputy director general,  
Confederation of Indian Industry (CII)

thin-film solar panels.

At the same time, the government needs to step up incentives and research funding so that the solar cells made in India can compete aggressively in terms of cost with those manufactured in, say, China. One step in this direction is reduction in the capital costs on solar cells by the Central Electricity Regulation Committee (CERC), which is the major reason for falling tariffs.

As tariffs have fallen significantly in the latest projects on offer and the trend is likely to continue, India has advanced the target date for selling solar power at the same rate as conventional electricity by five years to 2017. Companies have bid tariffs as low as Rs 7.49 per unit for solar plants, while tariff from existing coal projects is around Rs 5.50 per unit to Rs 6 per unit.

In the entire solar PV value chain, Indian companies are active only in cell and module manufacturing with no vertical integration. This makes their cells and modules costlier than products from Chinese counterparts.

Encouraging an entire PV ecosystem instead of just solar PV power plants is likely to result in cost reduction. India is still at the developing stage in terms of the availability of raw materials. The setting up of polysilicon plants will generate byproducts like silane gas, which can be used in production of solar cells. Currently, silane and other chemi-

cals are being imported.

Maharishi Solar seems to be the only company in India which has set up a vertically integrated manufacturing facility to produce multi-crystalline silicon ingots, multi-crystalline wafers, multi/mono solar cells, solar PV modules and solar PV systems at Srikalahasti, Andhra Pradesh.

Thankfully, successive reforms in the power sector and a plethora of policies initiated at the central and state levels to control greenhouse gas emissions and promote renewable energy have restored investors' interest in the solar power industry. The Indian solar market can look forward to large-scale private investments across the entire PV value chain, especially in the production of polysilicon feedstock, silicon wafers, PV modules and cells, as well as balance-of-system components.

High temperatures in many parts of India dictate better efficiency rates through use of thin-film solar cells. Currently, very few companies including Moser Baer, HHV Solar and Shurjo use thin-film technology. Work on thin-film technology needs to be carried out in India, as it will bring price competitiveness. The technology can be researched and developed to increase efficiency as well. Home-grown technology would definitely lower the cost of solar cells made in India. ●

The author is an executive editor at EFY